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CSE6242 / CX4242 Group Project Description

Group Project

CSE6242 / CX4242 Data and Visual Analytics

Deliverables

Proposal Deliverables

1. Proposal Document (7 course grade points)
2. Proposal Presentation (5)
3. Proposal Peer Feedback (1)

Midpoint Deliverables

4. Progress Report (5)
5. Midpoint Peer Feedback (2)

Final Deliverables

6. Final Poster Presentation (7)
7. Final Report (21)

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consuming to find "missing" files or to guess their names.

For example, suppose the final report requires README.txt and report.pdf; if your team submits README.doc and report.doc, 10% will be deducted from the final report's score.

Important: all policies stated in our syllabus (e.g., plagiarism, collaboration, timing) apply to all aspects of this course, including this project, unless specified otherwise. For example, as you are responsible for completing your own work, using text-to-speech tools to generate voice narration for presentations is clearly not allowed.

Teaming

The work will be carried out in teams of **4-6 people**. You are welcome to decide on who to team with since each team needs to decide on their topic.

Before the team formation deadline, we will share a spreadsheet and announce its availability. All teams will enter their team members' information there, and team numbers (IDs) will be automatically assigned. There is no need to inform us of your team member lineup ahead of time.

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grads and undergrads have different expectations and work schedules. If you want to form a group with both grads and undergrads:

1. We will grade that group as if **all members are grad students**
2. Every member **MUST** fully understand the potential challenges in coordinating work schedules (e.g., grads usually take classes on TH, undergrads on MWF) and expectations (e.g., course grades are generally very important for undergrads)

We will grade grad projects and undergrad projects separately; we generally expect grad projects to include more detailed analysis, comprehensive results, etc.

Why solo projects aren't offered?
The main reasons are: (1) most large-scale data analysis projects in the industry are team-based; (2) many former students found the projects highly beneficial to them. If you strongly desire to do solo projects, this course unfortunately is not a good fit for you.

Some students mistakenly believe that group projects reduce the amount of work that needs to be graded. Any instructors or TAs know that open-ended questions, and in this case projects with topics chosen by students, need a lot of thinking and time from the graders' end. In fact, an open-topic group project is one of the hardest things to scale to large classes. I thought hard about whether to remove it and do exams instead, which could have really saved us a lot of our time (TAs and myself)! I decided to keep the group project because of its benefits.

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your team and scope of work. Explicit requirements based on the number of team members such as the number of papers in your literature survey can be adjusted accordingly. Should the project scope be significantly and negatively affected due to the team member's departure, in the Appendix of the relevant documents (depending on which deliverables are affected: proposal, progress report, final report), describe what happened, the specific changes to the project scope, and compelling justifications for those changes. Teams that fall below the minimum size due to team members dropping may continue as long as the team agrees that they will be able to complete the planned tasks.

If you are planning to drop the course, please communicate your intention to your teammates as soon as possible so they can proceed accordingly! Your team members should proceed and re-distribute your work among the remaining members. If you decide to continue to support your team, you can assume an "advisor" role who may provide feedback, but you should not be writing code, etc. It's also important to set the right expectation or your teammates could unintentionally expect more from you than you want.

This course has many sections. How should students form teams from different sections?

The sections of this course are taught as the following two "cohorts":

- CSE6242A/CX4242A
- CSE6242OAN,O01,O3,Q

Students in the same cohort can form a team. For example, CSE6242OAN (OMSA) students can form a team with CSE6242O01 (OMSCS) students.

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Can we form a 7-student team?
We do not recommend forming a 7-student team, because of the likelihood of team challenges. However, we will allow it if all team members confirm that they have carefully reviewed and fully understood the potential challenges working in a larger 7-person team:

1. the potential challenges working in larger teams; and
2. Prof. Randy Pausch's advice on how to work successfully in a group. This is **VERY IMPORTANT**, especially if you have not worked together before.

To request approval for a 7-student team, send us a private Ed Discussion post (check the “Private” box) and include the exact statement “Our team confirms that all team members are fully aware of the potential challenges of working in a larger 7-person team, as mentioned on the project description page.”

Choosing a Topic

- **Pick your own topic:**
  - You need to justify that the topic is interesting, relevant to the course, and is of suitable difficulty.
  - **Three major required components:**
    1. At least one **large, real, public dataset** that your team can **readily access and use**. While APIs and web scraping might provide possible data access, they **will** significantly limit the

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2. Some **non-trivial** analysis/algorithms/computation performed on the dataset (e.g., computing basic statistics, like average, min/max will not be enough); and
3. An **interactive user interface** that interacts with the algorithms; the interface must be visual (and may additionally include other modalities, e.g., voice-controlled).

- **IMPORTANT**: Your team's project is work owned by your team members. As your team decides on the project topic, please be mindful that **the class project is NOT the best venue to express personal or subjective opinions** — rather, it is for students to practice performing objective analysis based on facts and science.
- This project **cannot** be related to any team member's existing or ongoing work, such as another course project a team member is working on, or a team member's research project or thesis. In other words, no "double-dipping" is allowed. This ensures there are meaningful and adequate opportunities for all team members to learn and contribute to the project's success.

Once you have selected a topic, you should do some background reading so that you are capable of describing, in some detail, what you expect to accomplish. For example, if you decide that you want to implement some new proposal for a multidimensional file structure, you will have to carefully read the paper that proposes similar structures, pinpoint their weaknesses, and explain how your approach will address these

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could be another person's "small". For example, in my research group, we routinely work with million-edge graphs (say, just working with the graph structure; you'll work on such "large" graphs in HW3 using just your computer), which are considered "small" in the data mining community, but large in other communities and applications. If you're working with videos, a few million of them will take up terabytes of petabytes. For those of you working in industry, you likely would routinely work with datasets that are in terabytes or petabytes.

The main reason for requiring the use of a large dataset is so that you will learn to handle non-trivial computing and visualization problems. So the larger the better. The harder the problem, the more thinking you will need to do, and the more you will learn.

If you can run an algorithm or analysis on your computer and get the results in a few seconds, your dataset is likely too small (and you likely won't learn much from this experience). Similarly, if you can plot every single data point on the screen trivially and that doesn't create any visual complexity or interaction challenges, your data is also likely too small. In other words, your team should encounter enough challenges to make yourselves consider the obstacles and strategies for tackling them!

If you have a large dataset and that makes the project too hard, you can always choose to work on a subset of it. But if your dataset is too small (e.g., a few hundreds of rows, each having only a few attributes), you will learn little.

I encourage you to pick an interesting topic and dataset (instead of a "safe" but boring topic) that would excite you -- this way, you would learn more. Be ambitious. It's OK if you end up getting negative results, as long as you make

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least hundreds of thousands of rows/records, or at least hundreds of MBs (however, if that mostly contains "filler" information that you won't be using, then that's not a meaningful measure). Again, the larger the better. Use this project as a way to gain experience and knowledge in working with real datasets.

Can we see example project deliverables from previous courses?
Unfortunately, I do not have permission to share previous project teams' deliverables. Also, since all teams are welcome to choose topics most interesting to them, different teams' ideas and approaches can be quite different --- there are many different ways to produce good proposals. Based on the project description and guidelines, most teams in the past developed excellent projects. In academia, when submitting grant proposals, we are not provided with any examples. It is up to us to propose the topic, and to convince the proposal reviewers the significance of our problems, ideas and solutions. We are only provided with high-level format requirements and guidelines of our documents.

Below are two published articles that are based on previous projects from this course (campus section), the articles themselves are not project deliverables from this course, but are like extended, improved version of the teams' final project reports. For our OMS students, these projects are mentioned in *Week 1's "Course Introduction: Course Goals & Expectations" video, start at 4:16*.

Aurigo: An Interactive Tour Planner for Personalized Itineraries

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PASSAGE, the students improved their methods of computing "safety" scores.

Proposal Document

Your proposal should answer Heilmeier's questions (all 9 of them; see list below); if you think a question is not very relevant, briefly explain why. In other words, your proposal should describe what you plan to do (the problem to address), why you want to do it, how you will do it (what tools? e.g., SQLite, PostgreSQL, Hadoop, Kinect, iPad, etc.), how your approach is better than the state of the art, why it may succeed, and when it does, what differences will it make, how you will measure success, how long it's gonna take, etc.

9 Heilmeier questions (source)

1. What are you trying to do? Articulate your objectives using absolutely no jargon.
2. How is it done today; what are the limits of current practice?
3. What's new in your approach? Why will it be successful?
4. Who cares?
5. If you're successful, what difference and impact will it make, and how do you measure them (e.g., via user studies, experiments, ground truth data, etc.)?
6. What are the risks and payoffs?
7. How much will it cost?

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else — including the literature survey — counts. **Use at least 1-inch margin for each page (top, right, bottom, left).** It must use **11pt font (or larger)**. The document must be in **PDF format. You may create the document using any software that you want; we highly recommend using LaTeX (if you are new to LaTeX, see below for a starter template).** Include any figures, charts, tables, captions, etc. whenever useful — they count towards the page limit (they may include text whose font size is smaller than 11pt, but such text must be legible). Your document should be self-contained. For example, do not just say: "We plan to implement Smith's Foo-Tree data structure [Smith86], and we will study its performance." Instead, you should briefly review the key ideas in the references, and describe clearly the alternatives that you will be examining.

Grading scheme & Submission instructions

- [60%] Literature survey
    - Your literature survey should have **at least 3** papers or book chapters per group member (outside of any required reading for the class).
        - Short papers, like PNAS, Nature, Science papers, count as 0.5.
        - Copying the abstract of the papers is obviously prohibited, constituting plagiarism.
        - For each paper, describe
            - (a) the main idea,
            - (b) why (or why not) it will be

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(needed if your team is using LaTeX).

- Make sure to cite your references in your literature survey.
- The literature survey can be in its own section, or be integrated into the answers of relevant Heilmeier questions (e.g., #2 and #3).

- [30%] Expected innovations
  - The project should be novel and exciting in the solutions for both areas of (1) algorithm/computation and (2) visualization.
- [10%] Plan of activities
  - Using either a **Gantt chart** (example) or a **table**, describe
    - the activities each member has done and will do; and
    - each activity's start and end time (or start time and duration).
  - [-5% if not included] Provide an **accurate statement** that summarizes the distribution of team members' effort. An accurate statement can be short, e.g., "all team members have contributed a similar amount of effort." Place this statement immediately after the Gantt chart (or table). If effort distribution is too uneven, we may assign higher scores to members who have contributed more.
- [-5%] For every Heilmeier question that's **not** mentioned.

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- Include your team project's title, team number, and all team member names (we recommend at the top of the first page)
- Team's contact person submits a softcopy, named **teamXXXproposal.pdf** (i.e., that person submits for the whole team), where XXX is the team number (e.g., team001proposal.pdf for team 1). Visit Canvas for submission instructions.

Extremely Important: all communication (e.g., emails, Teams/Slack messages) between team members **MUST be visible to all other team members at all times** — do not send private messages between only a subset of the team members. A great majority of the teams that had team conflicts violated this rule. All team members must read: https://www.cs.cmu.edu/~pausch/Randy/tipoForGroups.html

An appendix is for optional, non-essential information. We do not grade it. Thus, do NOT put your literature survey in an appendix.

Some teams, especially those aiming for a research publication, use LaTeX for formatting. If your team chooses this route, consider collaborative tools like Git (GT GitHub) or Overleaf. Georgia Tech offers free access to Overleaf Pro. You can sign up or link an existing Overleaf account. As long as all formatting requirements are met (e.g., at least 11-point font), your team can adapt any LaTeX template (e.g., start with an ACM template). Here is

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- See other articles related work sections for inspiration, e.g., Apolo paper
- Multiple papers may share similar themes, use similar methods so they may be summarized and discussed together.
- Note that the literature survey accounts for 60% of the proposal's grade, so your literature survey should be substantial!

Which papers are considered "long" (or "short")?
Long papers refer to typical papers published at top academic venues (e.g., KDD, CHI, ICML). They are usually at least 8-10 pages long, in 2-column format, which translate into 5000 or more words. Thus, short paper would be 4-5 pages or fewer. Example long papers:

- GAN Lab. VAST'18. 2-column IEEE format
- SHIELD. KDD'18. 2-column ACM format
- ShapeShifter. PKDD'18, 1-column Springer format

Should papers be peer-reviewed? How to tell if a paper was peer-reviewed?
Yes, they should be peer-reviewed, unless there is a strong reason for it not to be (e.g., a book chapter).

You can usually find out whether a journal or conference proceeding is peer-reviewed by checking its submission and reviewing process (or lack thereof), by visiting the conferences or publishers' websites. A quick way to look up the venue at where a paper was published is to plug in the paper's title into Google Scholar. A **very short** of these for data analytics/Machine Learning/AI include: KDD, IEEE ICDM, IEEE Vis, UbiComp, SIAM SDM, ICDM, NeurIPS, ICML, AAAI, IJCAI, AISTATS, and many more.

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GT Library or repeatedly sign in via GT authorizations on multiple publishers' sites.

What kind of papers are considered relevant?
A paper that you read and cite can be relevant to your project in different ways. You are welcome to cite a paper if you can justify its strong relevance to your ideas, problems (e.g., motivate the urgent need to solve them), or approaches (e.g., your approach improves on an existing method). Searching on Google Scholar can also help you to find relevant papers.

Can we use tools like ChatGPT for data processing and visualization?
The short answer is "no" if the goal of it is to avoid thinking through the challenges in data processing and visualization and coming up with solutions for them. A good way to think about this: if that tool isn't available, will your team still be able to tell what sorts of data processing and visualization are necessary, why they are necessary, how they are done, etc?

Proposal Presentation

1. [OMS and campus] Team's contact person submits a presentation slide deck called **teamXXXslides.pdf**, where XXX is the team number (e.g., team001slides.pdf

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on your computer screen via screen capture, say using Quicktime, MonoSnap, etc.) with voice narration; it is up to you whether to show your face. You should be able to create this recording quickly with little effort – no need to do any special video or audio editing.

2.2. Upload the video as an **unlisted YouTube video** (NOT "private" or "public"). Unlisted videos can be viewed by anyone with the link to your unlisted video.

2.3. Submit the **URL (web link)** of the unlisted YouTube video. The graders will use this URL to view the video. To double-check that the URL works, visit that URL using a separate web browser that has been fully logged out of Google services (e.g., all cache cleared, use "Incognito" mode in Chrome, etc.). Visit Canvas for submission instructions.

2.4. Set the title of the YouTube video to **teamXXXproposal**, where XXX is the team number (e.g., 001 for team 1).

2.5. **IMPORTANT**: you need a Google Account to upload a video to YouTube. To access YouTube, depending on your geographic location, you may need to use VPN (e.g., Georgia Tech's VPN). **Uploading a large video file can take a lot of time; VPN can further slow that down.** Make sure you finish creating and uploading your video early, so you have ample time to verify that your submission is successful.

3. [Campus only]

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3.2. If you overrun, **we will boot you off the stage and deduct 5% of your presentation grade.**

3.3. Two teams can swap their presentation dates if both teams agree; no need to ask Polo for permission. Both teams MUST write to us via an Ed Discussion private post at least 48 hours before the first proposal presentation class time, to request that the spreadsheet be updated with the new dates.

3.4. **Attendance is mandatory.** Lateness and absence will be noted; marks may be deducted for no-show. If you cannot attend, let Polo and TAs know through a private post on Ed Discussion. We want you all to attend and support your peers, whether they are part of your own team or another. Observing how others present helps you improve your own presentation skills. Additionally, showing support for teams presenting before or after yours is common courtesy and a mark of respect.

3.5. Every team's presentation slides will be collected via Canvas a few days before the proposal days, and pre-loaded to the computer at the podium, to reduce switching time between teams.

Grading

- [45%] You must answer the Heilmeier questions. 5% for each question. If a question doesn’t apply, say so.

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    - The project should be novel and exciting in the solutions for both areas of (1) algorithm/computation and (2) visualization.
- [10%] Plan of activities (same requirements as described in “Proposal” section above)
- [20%] Presentation delivery
- [-5%] Illegible text, tiny figures, bad color contrast, etc.
- [-5%] Overrun. In addition, you may lose additional points for required content that is not covered within the time limit.
- Your presentation does NOT need to strictly follow your project proposal document. For example, you can talk about ideas and materials that your team has come up recently.
- **Points will NOT be deducted or awarded based on the number of presenters.** We saw great presentations delivered by teams having various numbers of presenters.

Tips

- Use few slides. Less is more! Fewer slides mean less likely to overrun. Being succinct is hard.
- Practice timing and delivery! If you have several speakers, make sure you practice how to transition from one person to the next (e.g., passing the mic, passing control of mouse and keyboard, etc). PRACTICE! PRACTICE! PRACTICE!

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everyone is contributing to the project, each student will complete three anonymous peer feedback surveys on Canvas:

- **Proposal Peer Feedback**: covers participation up to and including proposal submission
- **Midpoint Peer Feedback**: covers participation from after proposal submission up to and including progress report submission
- **Final Peer Feedback**: covers participation after progress report submission

You will provide anonymous feedback on each teammate. For example, in a 6-person team, you will complete 5 feedback tasks. Be candid and respectful. Your feedback helps teammates understand what's working, identify communication or participation issues, and reflect on how to improve. Each survey is graded for completion.

Course staff will review results. By default, feedback will **not** be used to adjust any project scores. However, if a team requests our help to resolve team issues they cannot handle on their own, we may review that team's survey results more closely to support a resolution.

More information and detailed instructions will be provided before the survey opens.

Progress Report

No more than **4 letter-size pages** (excluding references)**, 11pt font, typed**. **Use at least 1-inch margin for each page (top, right, bottom, left).**

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project.

Specifically, the introduction and literature survey sections should be in their final form. The section on the proposed method should be almost finished. The sections about experiments and conclusions will have whatever results you have obtained, as well as “place-holders” for the results you plan/hope to obtain.

Grading scheme & Submission instructions

- [70%] Detailed description of your team’s methods (should be almost finished)
    - Provide a clear **list of innovations**: give a list of the best 2-4 ideas that your approach exhibits.
    - This section should be almost finished. That is, we expect your team to have made substantial progress in the design and implementation of BOTH the computation and visualization components.
- [25%] Detailed description of the design of upcoming experiments for evaluating your approaches
- [5%] Plan of activities (same requirements as described in “Proposal” above)
    - Provide the old plan and the revised plan
    - [-5% if not included] Provide an **accurate statement** that summarizes the distribution of team members’ effort. An accurate statement can be short, e.g., "all team members have contributed a similar amount of effort." Place this statement immediately after the Gantt chart (or table). If effort distribution is too uneven, we may assign higher scores to members who have contributed more.

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team 1). Visit Canvas for submission instructions.

An appendix is for optional, non-essential information. We may not read or even grade it.

Do we need to explicitly answer all Heilmeier questions in the Progress Report?
You are not required to explicitly answer the Heilmeier questions in the Progress Report like you did in the proposal. The main purpose of the Heilmeier questions (at the proposal stage) was to help your team think through relevant parts of the problem being studied and the proposed solutions, to figure out a plan of what to do. As you likely have already figured out, some content from the proposal is still very much relevant (e.g., the main ideas and possible technical approaches) and you team may likely want to expand on such content. And some content may need to be modified or even removed.

The progress report may be written based on your proposal. For example, the literature survey in the progress report is not required to be identical to the literature survey in the proposal. That is, you may update the proposal's literature survey as needed. Of course, the number of papers should not drop below the requirement (3 papers/team member), and the quality of discussion should still be equal or better than that in the proposal.

What do you mean by “evaluation”?
To evaluate your team’s proposed approaches, you come up with experiments to test how well they address the

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The type of experiments depend on your proposed approach(es) and problem(s). Suppose a team is proposing an movie recommendation tool that allows users to interactively specify movies that they like and receive recommendations in real time, then possible experiments may include:

1) Scalability evaluation, e.g., how well the recommendation algorithm scales with data (e.g., wall clock time v.s. #movies in dataset)

2) Recommendation accuracy, e.g., using MovieLens datasets

3) Usability evaluation of the tool, e.g., via user study (for a class project, it's acceptable to invite classmates; for research that you plan to publish, you will need to go through the formal IRB protocol process)

Final Poster Presentation [Peer-graded]

Overview

1. Each team creates a single poster for the whole team.
2. Each team member separately prepares and creates a **3-minute** video presentation (i.e., one presentation per learner).

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narration (e.g., as pdf on your computer screen via screen capture, say using MonoSnap, native screen recording software on your OS). It is up to you whether to show your face. You should be able to create this recording quickly with little effort – no need to do any special video or audio editing. You may zoom into and out of the poster as you present, so the viewer can more easily see the poster content.

2.3. Demo: optional but encouraged. Demo time counts towards presentation time.

3. Upload your video as an **unlisted YouTube video** (NOT "private" or "public"). Unlisted videos can be viewed by anyone (in this case, peer-graders who grade your presentation) with the link to your unlisted video.

   3.1. Submit the **URL (web link)** of your own unlisted YouTube video via Canvas. Your graders will use this URL to view your video. To double-check that your URL works, visit that URL using a separate web browser that has been fully logged out of Google services (e.g., all cache cleared, use "Incognito" mode in Chrome, etc.)

   3.2. Set the title of your YouTube video to **teamXXXposter-YY**, where XXX is the team number (e.g., 001 for team 1), and YY is the student's last name (e.g., smith). It is OK if multiple members have the same last name, because videos are uniquely identified by their URLs.

   3.3. **IMPORTANT**: you need a Google Account to upload a video to YouTube. To access YouTube, depending on your geographic location, you may need to use VPN (e.g., Georgia Tech's VPN). **Uploading a large video file can take a**

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4.1. **Peer grading is NOT anonymous.** That is, a presenter knows who the graders are, and a grader knows who the presenters are.

5. If a grader does not finish all the assigned peer grading, that grader **may NOT receive all or part of the grader's own final poster presentation grade** (i.e., up to 7.5% of final course grade), since the peer grading is an integral part of the project presentation.

6. We will compute a student's final poster presentation score as follows. For each rubric item, we take the average of the two highest scores. Then, we sum up these "averages" as the student's final score. This formula should heavily suppress "outlier" scores. After the peer grading ends, as additional safeguard, we will go through everyone's scores.

For example, suppose
for "What is the problem(no jargon)?", the student receives 4, 2, 3; and
for "Why is it important and why should we care?", the student receives 4, 4, 5;
the final score is computed as:

```
final score = sum(avg(4, 3) + avg(4, 5)
+ ... )
```

where "4, 3" are the two highest scores from "4, 2, 3"; and "4, 5" are the two highest from "4, 4, 5".

Why unlisted YouTube video?

In previous semesters, students submitted video files via Canvas. That did not work well, creating significant

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People rarely like to hear about negative comments, even if they are accurate and helpful. Giving negative news is always hard, but that is part of life! This means we should carefully phrase our comments as constructive criticism. For example,

- instead of saying “too much text and not enough figures”, you could say "Fig 1 to 3 are important figures in this project; currently they are not easy to see (images are too small; text is not legible). Suggest reducing the amount of text, e.g., into succinct, bullet points to create space for the figures".
- Similarly, avoid "I don't think that the visualization is anything new or how it is helpful," which is highly subjective. Instead, justify your comments; if the presenter did not clarify the novelty or significance of an approach (it is probably new, but just that the presenter did not point it out), you could say "it's unclear from the presentation and poster whether the proposed visualization is an improvement over the state of the art (it seems to be a standard design); more clarification is needed."

There are pros and cons for both anonymous and open review. It is still an open research problem. For example, one potential benefit for open review is reviewers could be more tactful and constructive, where anonymous reviewers could be more critical (sometimes not in a good way) and may do less work than they should.

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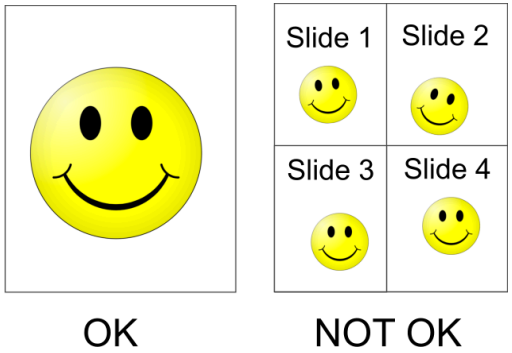
The poster must be in portrait orientation, **36 inches wide and 40 inches tall**. We suggest using **18pt** font size and larger.

A deck of PowerPoint slides is **not acceptable** as a poster. See the illustration below for what is allowed and what is not.



Grading

Your poster presentation should cover the following parts (point distribution shown on the left). Thus, the grading is about both your presentation delivery (e.g., what you say, where you direct the audience's attention), and the poster content.

If you overrun, besides losing points for the rubric item "5% Finished on time?", you may lose additional points for the required content that you have not covered within the time limit --- imagine you are delivering a presentation in person and you are alloted 3 minutes, once that time is up, you would need to stop and would not be able to present additional content (thus, that content will not be graded).

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	5% How do they work? 5% Why do you think they can effectively solve your problem (i.e., what is the intuition behind your approaches)? 5% What is new in your approaches?
10%	**Data**: 5% How did you get it? (Download? Scrape?) 5% What are its characteristics (e.g., size on disk, # of records, temporal or not, etc.)
25%	**Experiments and results**: 5% How did you evaluate your approaches? 10% What are the results? 10% How do your methods compare to other methods?
10%	**Presentation delivery**: 5% Finished on time? 5% Spoke clearly and at a good pace?
25%	**Poster Design**: 5% Layout/organization (Clear headings? Easy to follow?) 5% Use of text (Succinct or verbose?) 5% Use of graphics (Are they relevant? Do they help you better understand the project's approaches and ideas?) 5% Legibility (Is the text and figures too small?) 5% Grammar and spelling

How should you grade it if audio narration is missing?

Please grade using your best judgment based only on what you can see (and not hear). For example, certain rubric items would likely receive very low scores (e.g., “Spoke clearly and at a good pace?”), and some rubric items could become unsatisfactory, confusing, or even “missing” because a voice narration might have helped direct the

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included on the poster but not mentioned in the narration?

This depends on how well they coordinate what they say and what they show on the poster. For example, assuming the "extreme" scenario where the information is shown in a very small font, is never mentioned, and wouldn't even be visible unless you pause the video, then a low score should probably be given (e.g., "missing" or "confusing"). Otherwise, you could imagine a poster simply packing the whole final report in, placing the onus on the grader to "discover" what's there and what's not.

Possible software to create posters

1. Figma -- free for students (Polo highly recommends)
2. Powerpoint/Word (save as pdf) -- GT's Office365 Powerpoint supports collaboration.
3. Apple Pages (FREE) supports real-time collaboration (via iCloud and desktop software)
4. Inkscape (free, cross platform)
5. Polo uses Affinity Designer (Mac and windows)
6. Google Drawings (File > Page Setup to set document size)
7. draw.io (File > Page Setup to set document size)

Example poster design

The following posters were for research projects conducted at the Polo Club of Data Science, and were not for projects from this class. They do not strictly follow the format described in our grading rubric.

- Apolo graph exploration
- Insider trading pattern discovery
- Comment spam detection

Where to print posters on Atlanta campus? (Not applicable this semester)

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Final Report

It will be a detailed description of what you did, what results you obtained, and what you have learned and/or can conclude from your work.

Components:

1. **Writeup**: no more than **6 letter-size pages (excluding references), 11pt font, typed**. **Use at least 1-inch margin for each page (top, right, bottom, left).** Describe in depth the novelties of your approach and your discoveries/insights/experiments, etc.
2. **Software**: packaging, documentation, and portability. The goal is to provide enough material, so that other people can use it and continue your work, if you are to open-source it --- in other words, you should make it easy and attractive for others to use your work.

Grading scheme & Submission instructions

- Writeup, with six sections numbered and named exactly as follows (highlighted in bold font)
    - **1. Introduction** [2%] — motivation
    - **2. Problem Definition** [3%] — provide a precise formal problem definition, in addition to the

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    2. [35%] Detailed description of your approaches: algorithms, user interfaces, etc.

- **5. Evaluation**
    1. [5%] Description of your testbed; list of questions your experiments are designed to answer
    2. [25%] Detailed description of the experiments; observations (as **many** as you can!)
- **6. Conclusions and Discussion** [5%]
    1. Depending on the topic of your project, this section may clearly summarize the key points of the project, such as the main ideas, results, impacts, significances, etc., and discuss issues such as limitations, implications, potential future extensions, etc.
    2. [-5% if not included] Provide an **accurate statement** that summarizes the distribution of team members' effort. An accurate statement can be short, e.g., "all team members have contributed a similar amount of effort." If effort distribution is too uneven, we may assign higher scores to members who have contributed more.
- Include your team project's title, team number, and all team member names (we recommend at the top of the first page)

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corresponding to the "user guide". This file should contain:

1. DESCRIPTION - Describe the package in a few paragraphs
2. INSTALLATION - How to install and setup your code
3. EXECUTION - How to run a demo on your code
4. [Optional, but recommended] DEMO VIDEO - Include the URL of a 1-minute ***unlisted* YouTube video** in this txt file. The video would show how to install and execute your system/tool/approach (e.g, from typing the first command to compile, to system launching, and running some examples). Feel free to speed up the video if needed (e.g., remove less relevant video segments). This video is optional (i.e., submitting a video does not increase scores; not submitting one does not decrease scores). However, we recommend teams to try and create such a video, because making the video helps teams better think through what they may want to write in the README.txt, and generally how they want to "sell" their work.

- **DOC** - a folder called DOC (short for “documentation”) containing:
  1. **teamXXXreport.pdf** - Your report writeup in PDF format; can be created

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Should datasets be included as part of our submission?

If you are referring to (small) toy data for a demo (that your graders will run), you are welcome to include them. Think about the open-source software libraries that you have seen or have used, they would often include some sort of "quick start" guide to get a demo running on a toy dataset.

For large datasets, please do **not** include them; if the dataset is public and can be easily downloaded, include the link to the dataset.

If getting a dataset requires writing scripts/programs, include those scripts, and write down the steps that people will need to go through (e.g., register for an account to get API key).

If you have processed the dataset in some ways, include the code you used, and the steps people will need to go through.

Can my team publish our project (e.g., on GitHub)?

Your team collectively owns the work. Discuss with your teammates before making anything public, and if your team decides to do so, ensure it is done in a manner unlikely to encourage future plagiarism. For instance, you may open-source the project similar to how typical open-source projects would. Definitely, do not simply upload deliverables that your group developed specifically for this course (e.g., proposal.pdf, proposal.docx, etc.).

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apply for the Startup Launch Create-X in Georgia Tech. If your team works on a good idea and you would like to take the next step and commercialize your idea and start your own company, I strongly recommend to apply for the Startup Launch in Create-X and take advantage of Funding, Coaching and many more supports that come with this program.

Updated:

2025-12-30